Discrete Structures - Abstract Algebra

Lecture Notes on Groups and Rings

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Monsoon 2026

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Sources. Each subsection begins with a *Reading* line that points to the matching chapter in J. A. Gallian, *Contemporary Abstract Algebra*, and the matching section in I. N. Herstein, *Topics in Algebra* (2nd edition). Each section ends with exercises and a pointer to the problem sets of the two books. Page and theorem numbers are not given, because they differ between editions. The homomorphism section follows the topics of K. Conrad's handout *Homomorphisms*.

1 Groups

1.1 Binary Operations

Reading. Gallian, Ch. 2 (Groups); Herstein, §2.1.

Definition 1.1 (Binary operation)**.** Let G be a set. A *binary operation* on G is a rule $\cdot : G \times G \to G$ that assigns to every ordered pair (a, b) of elements of G an element $a \cdot b$ of G.

The important point is that $a \cdot b$ again lies in G for all $a, b \in G$. This property is called *closure*: the operation produces an element that is already in G.

Example 1.2. (1) $G = \mathbb{Z}$ and $\cdot = +$. For all $a, b \in \mathbb{Z}$ we have $a + b \in \mathbb{Z}$.

(2) $G = \mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ and $\cdot$ is addition modulo n.

1.2 The Definition of a Group

Reading. Gallian, Ch. 2; Herstein, §2.1.

Definition 1.3 (Group)**.** Let G be a set with a binary operation $\cdot$. We say that $(G, \cdot)$ is a *group* if the following three properties hold.

(G1) **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in G$.

(G2) **Identity:** There is an element $e \in G$ such that $a \cdot e = e \cdot a = a$ for all $a \in G$.

(G3) **Inverses:** For every $a \in G$ there is an element $b \in G$ such that $a \cdot b = b \cdot a = e$.

Remark 1.4. (1) We will soon drop the symbol "$\cdot$" and write ab instead of $a \cdot b$.

(2) It is not always true that $ab = ba$. If $ab = ba$ for all $a, b \in G$, then G is called *abelian*.

1.3 Examples and Non-Examples

Reading. Gallian, Ch. 2 and Ch. 3 (Examples); Herstein, §2.2.

Proposition 1.5. *Each of the following is a group.*

(1) $\mathbb{Z}$, $\mathbb{Q}$ *and* $\mathbb{R}$ *under addition.*

(2) The set $\{1, -1, i, -i\} \subseteq \mathbb{C}$ *under complex multiplication.*

(3) $\mathbb{Q}^+$*, the positive rational numbers, under ordinary multiplication.*

(4) $\mathbb{R}^*$*, the set of non-zero real numbers, under ordinary multiplication.*

(5) The set of all 2×2 *matrices with real entries under entry-wise addition.*

(6) For $n \geqslant 1$*, the set* $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ *under addition modulo* n*.*

Proof. (1) Addition of numbers is associative. The identity is 0 and the inverse of a is $-a$. The sum of two integers (rationals, reals) is again an integer (rational, real).

(2) Write $i^0 = 1$, $i^1 = i$, $i^2 = -1$, $i^3 = -i$. The set is $\{i^0, i^1, i^2, i^3\}$, and $i^4 = 1$. For $0 \leqslant s, t \leqslant 3$ we have $i^s i^t = i^{s+t} = i^{(s+t) \bmod 4}$, which lies in the set. So the set is closed. Associativity is inherited from complex multiplication. The identity is $i^0 = 1$. The inverse of i^s is i^{4-s}, because $i^s i^{4-s} = i^4 = 1$.

(3) The product of two positive rational numbers is a positive rational number. Multiplication of numbers is associative. The identity is 1 and the inverse of a is $1/a$, which is again positive and rational.

(4) If $a \neq 0$ and $b \neq 0$ then $ab \neq 0$, so the set is closed. Multiplication is associative, the identity is 1, and the inverse of a is $1/a \in \mathbb{R}^*$.

(5) The sum of two 2×2 real matrices is a 2×2 real matrix. Entry-wise addition is associative because addition in $\mathbb{R}$ is. The identity is the zero matrix $\left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$, and the inverse of a matrix is the matrix of negated entries.

(6) Write $a \oplus b$ for the remainder of $a + b$ on division by n; it lies in $\{0, \ldots, n-1\}$. Both $(a \oplus b) \oplus c$ and $a \oplus (b \oplus c)$ are numbers in $\{0, \ldots, n-1\}$ that are congruent to $a + b + c$ modulo n. Two such numbers must be equal, so $\oplus$ is associative. The identity is 0. The inverse of $j \neq 0$ is $n - j$, because $j + (n - j) = n \equiv 0 \pmod{n}$, and the inverse of 0 is 0. □

Non-example 1.6. (1) $\mathbb{Z}$ is not a group under multiplication. The identity would have to be 1, and there is no integer b with $2b = 1$ (as $2b$ is even). So 2 has no inverse; its inverse would be $\frac{1}{2} \notin \mathbb{Z}$.

(2) Let $S = \{\text{positive irrational numbers}\} \cup \{1\}$. Then S is not a group under multiplication, because $\sqrt{2} \cdot \sqrt{2} = 2 \notin S$. Closure fails.

(3) $\mathbb{Z}$ under subtraction is not a group, because associativity fails: $(5 - 3) - 2 = 0$ but $5 - (3 - 2) = 4$.

(4) The set of all 2×2 real matrices is not a group under matrix multiplication (see Example 1.7 below).

Example 1.7 (The general linear group)**.** Let

$$\mathrm{GL}(2, \mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \mathbb{R},\ ad - bc \neq 0 \right\}.$$

This is a group under matrix multiplication.

- *Closure.* Let $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} x & y \\ z & w \end{smallmatrix}\right]$. Then $AB = \left[\begin{smallmatrix} ax+bz & ay+bw \\ cx+dz & cy+dw \end{smallmatrix}\right]$ and

$$\begin{aligned} \det(AB) &= (ax + bz)(cy + dw) - (ay + bw)(cx + dz) \\ &= (acxy + adxw + bcyz + bdzw) - (acxy + adyz + bcxw + bdzw) \\ &= ad(xw - yz) - bc(xw - yz) = (ad - bc)(xw - yz) = \det A \det B. \end{aligned}$$

  So if $\det A \neq 0$ and $\det B \neq 0$, then $\det(AB) \neq 0$ and $AB \in \mathrm{GL}(2, \mathbb{R})$.

- *Associativity.* Matrix multiplication is associative. (Each entry of $(AB)C$ and of $A(BC)$ is the same sum of products $a_{ij} b_{jk} c_{kl}$.)

- *Identity.* The identity matrix $I = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ satisfies $IA = AI = A$ and has determinant 1.

- *Inverses.* Let $\Delta = ad - bc \neq 0$ and

$$A' = \begin{bmatrix} \dfrac{d}{\Delta} & \dfrac{-b}{\Delta} \\ \dfrac{-c}{\Delta} & \dfrac{a}{\Delta} \end{bmatrix}.$$

Then

$$AA' = \frac{1}{\Delta}\begin{bmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{bmatrix} = I,$$

and in the same way $A'A = I$. So $A' = A^{-1}$. Since $\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det I = 1$, we have $\det(A^{-1}) = \det(A)^{-1} \neq 0$, so $A^{-1} \in \mathrm{GL}(2, \mathbb{R})$.

Remark 1.8. (1) $\mathrm{GL}(2, \mathbb{R})$ is a *non-abelian* group. For example, with $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ we get $AB = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ but $BA = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$. The definition extends to arbitrary $n \times n$ matrices.

(2) If $ad - bc = 0$, then the matrix has no inverse. Indeed, if $AB = I$ for some B, then $\det A \det B = \det I = 1$ by the determinant formula above, which is impossible when $\det A = 0$. Hence the set of all 2×2 real matrices does not form a group under matrix multiplication.

1.4 Cayley Tables and the Group $U(n)$

Reading. Gallian, Ch. 2 and Ch. 3 ($U(n)$); Herstein, §2.2.

Definition 1.9 ($U(n)$)**.** For $n \geqslant 2$ let

$$U(n) = \{\, a : 1 \leqslant a < n,\ \gcd(a, n) = 1 \,\},$$

the set of positive integers less than n that are relatively prime to n.

Theorem 1.10. *$U(n)$ is a group under multiplication modulo n.*

Proof. Suppose $\gcd(a, n) = \gcd(b, n) = 1$. If $\gcd(ab, n) > 1$, some prime p would divide both ab and n. Then $p \mid a$ or $p \mid b$, so p would divide $\gcd(a, n)$ or $\gcd(b, n)$, a contradiction. So $\gcd(ab, n) = 1$. Reducing ab modulo n does not change its gcd with n, so $ab \bmod n$ again lies in $U(n)$. Associativity is inherited from multiplication of integers, and 1 is the identity. For inverses: if $\gcd(a, n) = 1$ then by Bézout's identity there are integers x, y with $ax + ny = 1$. Then $ax \equiv 1 \pmod{n}$, and $x \bmod n$ is an inverse of a that lies in $U(n)$. □

Example 1.11. For $n = 10$ we have $U(10) = \{1, 3, 7, 9\}$. Its multiplication table modulo 10 is

mod 10	1	3	7	9
1	1	3	7	9
3	3	9	1	7
7	7	1	9	3
9	9	7	3	1

For example, $7 \times 9 = 63 \equiv 3 \pmod{10}$.

Definition 1.12 (Cayley table)**.** A table that lists the result of the binary operation on all pairs of group elements is called a *Cayley table*.

Remark 1.13. If n is prime, then $U(n) = \{1, 2, \ldots, n - 1\}$.

1.5 Symmetries of a Square

Reading. Gallian, Ch. 1 (Symmetries of a Square) and Ch. 3 (dihedral groups); Herstein, §2.2.

Take a square whose four corners carry the labels P, W, G, B as shown. A *symmetry* of the square is a way to move the square so that it covers its original position. Each symmetry rearranges the four labels.

Name	Result	Meaning
(start)	P W G B	original square
R_0	P W G B	no movement
R_{90}	W B P G	rotation by 90° (anticlockwise)
R_{180}	B G W P	rotation by 180°
R_{270}	G P B W	rotation by 270° (anticlockwise)
H	G B P W	flip across the horizontal axis
V	W P B G	flip across the vertical axis
D	P G W B	flip across the main diagonal
D'	B W G P	flip across the other diagonal

Convention 1.14. We write XY for "first do Y, then do X" (the same order as composition of functions).

Example 1.15. Doing R_{90} and then H gives the same result as D. In symbols, $H R_{90} = D$.

Theorem 1.16. *The set* $D_4 = \{R_0, R_{90}, R_{180}, R_{270}, H, V, D, D'\}$ *is a group under composition of symmetries, and* D_4 *is not abelian.*

Proof. A symmetry is determined by where it sends each of the four corner positions of the square. So we can view each of the eight symmetries as a bijection of the set of four corner positions, and XY is the composition of bijections "first Y, then X". In this way the entries of the table above (which lists the rearranged labels) determine each symmetry, because the four labels are distinct.

Associativity. Composition of functions is always associative: $\big((X \circ Y) \circ Z\big)(p) = X(Y(Z(p))) = \big(X \circ (Y \circ Z)\big)(p)$.

Closure. Computing all 64 products (apply Y to the labelled square, then X) gives the following table, in which the entry in row X and column Y is XY. Every entry is one of the eight symmetries.

	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_0	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	R_0	D'	D	H	V
R_{180}	R_{180}	R_{270}	R_0	R_{90}	V	H	D'	D
R_{270}	R_{270}	R_0	R_{90}	R_{180}	D	D'	V	H
H	H	D	V	D'	R_0	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	R_0	R_{270}	R_{90}
D	D	V	D'	H	R_{270}	R_{90}	R_0	R_{180}
D'	D'	H	D	V	R_{90}	R_{270}	R_{180}	R_0

(As an example of how an entry is found: R_{90} turns $\boxed{\begin{matrix} P & W \\ G & B \end{matrix}}$ into $\boxed{\begin{matrix} W & B \\ P & G \end{matrix}}$, and then H turns this into $\boxed{\begin{matrix} P & G \\ W & B \end{matrix}}$, which is D. So $HR_{90} = D$, the entry in row H, column R_{90}.)

Identity. Row R_0 and column R_0 repeat the headings, so $R_0X = XR_0 = X$.

Inverses. The table has R_0 in row X and column Y exactly for the following pairs, and then $XY = YX = R_0$ (the table is symmetric about the main diagonal at these entries):

$$R_0^{-1} = R_0, \quad R_{90}^{-1} = R_{270}, \quad R_{180}^{-1} = R_{180}, \quad R_{270}^{-1} = R_{90},$$
$$H^{-1} = H, \quad V^{-1} = V, \quad D^{-1} = D, \quad (D')^{-1} = D'.$$

Not abelian. Row H, column D gives $HD = R_{90}$, while row D, column H gives $DH = R_{270}$. Since $R_{90} \neq R_{270}$, we have $HD \neq DH$. □

1.6 Preliminary Lemmas

Reading. Gallian, Ch. 2 (elementary properties of groups); Herstein, §2.3.

Lemma 1.17. *Let G be a group. Then:*

(0) ***(Cancellation)*** *If $ba = ca$ then $b = c$. Similarly, if $ab = ac$ then $b = c$.*

(1) The identity element e of G is unique.

(2) Every $a \in G$ has a unique inverse in G. We write it a^{-1}.

(3) For every $a \in G$, $(a^{-1})^{-1} = a$.

(4) For all $a, b \in G$, $(ab)^{-1} = b^{-1}a^{-1}$.

Proof. (0) Suppose $ba = ca$. Multiply on the right by a^{-1} and use associativity:

$$b = b(aa^{-1}) = (ba)a^{-1} = (ca)a^{-1} = c(aa^{-1}) = c.$$

If $ab = ac$, multiply on the left by a^{-1} in the same way.

(1) Let e and f both be identities. Since e is an identity, $ef = f$. Since f is an identity, $ef = e$. Hence $e = f$.

(2) Suppose x and y are both inverses of a. Then

$$x = ex = (ya)x = y(ax) = ye = y.$$

(3) We have $a^{-1}a = aa^{-1} = e$. So a is an inverse of a^{-1}. By (2) the inverse of a^{-1} is unique, so $(a^{-1})^{-1} = a$.

(4) Compute

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aa^{-1} = e, \qquad (b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = b^{-1}b = e.$$

So $b^{-1}a^{-1}$ is an inverse of ab. By (2), $(ab)^{-1} = b^{-1}a^{-1}$. □

Exercises

E1.1 Let G be a group with $a^2 = e$ for every $a \in G$. Prove that G is abelian.

E1.2 Let G be a group with $(ab)^2 = a^2b^2$ for all $a, b \in G$. Prove that G is abelian.

E1.3 Let S be a finite non-empty set with an associative operation such that $ab = ac \Rightarrow b = c$ and $ba = ca \Rightarrow b = c$ for all $a, b, c \in S$. Prove that S is a group.

E1.4 Prove that $SL_2(\mathbb{R}) = \{A \in GL_2(\mathbb{R}) : \det A = 1\}$ is a group under matrix multiplication.

E1.5 Write the Cayley table of $U(12)$. Show that $a^2 = 1$ for every $a \in U(12)$.

E1.6 Find all elements of D_4 that commute with every element of D_4.

E1.7 Let a, b be elements of a group. Prove that ab and ba have the same order.

E1.8 Prove that the unit circle $\{z \in \mathbb{C} : |z| = 1\}$ is a group under multiplication.

More problems. Gallian: Ch. 2 and Ch. 3, the exercise sets on groups, $U(n)$ and dihedral groups. Herstein: §2.1–§2.3, the problem sets at the end of each section.

2 Subgroups, Cosets, and Lagrange's Theorem

2.1 Subgroups

Reading. Gallian, Ch. 3 (Finite Groups; Subgroups); Herstein, §2.4.

Definition 2.1 (Subgroup)**.** A non-empty subset H of a group G is a *subgroup* of G if H is itself a group under the binary operation of G.

Lemma 2.2 (Subgroup test)**.** *A non-empty subset H of a group G is a subgroup of G if and only if*

(1) $a, b \in H \Rightarrow ab \in H$, *and*

(2) $a \in H \Rightarrow a^{-1} \in H$.

Proof. ($\Rightarrow$) Let H be a subgroup. Closure gives (1). Let e_H be the identity of H. Then $e_H e_H = e_H = e_H e$ in G, so $e_H = e$ by cancellation. If $a \in H$ and $b \in H$ is its inverse in H, then $ab = e$, so $b = a^{-1}$ by uniqueness of inverses in G. This gives (2).

($\Leftarrow$) Assume (1) and (2). Closure is (1). Associativity holds in H because it holds in G. Since $H \neq \emptyset$, pick $a \in H$. By (2) $a^{-1} \in H$, and by (1) $e = aa^{-1} \in H$, so H contains the identity. Inverses exist by (2). □

For finite sets, closure alone is enough.

Lemma 2.3 (Finite subsets)**.** *If H is a non-empty finite subset of a group G and H is closed under the operation of G, then H is a subgroup of G.*

Proof. Let $a \in H$. Because H is closed, all positive powers $a, a^2, a^3, \ldots$ belong to H. Since H is finite, these powers cannot all be different. So there are integers $r > s > 0$ with $a^r = a^s$. Cancelling a^s gives $a^{r-s} = e$. Hence $e = a^{r-s} \in H$ (note $r - s \geqslant 1$).

Now the element a^{r-s-1} lies in H: if $r - s - 1 \geqslant 1$ it is a positive power of a, and if $r - s - 1 = 0$ it equals $e \in H$. Also $a \cdot a^{r-s-1} = a^{r-s} = e$ and $a^{r-s-1}a = e$. So $a^{-1} = a^{r-s-1} \in H$. By Lemma 2.2, H is a subgroup. □

Corollary 2.4 (Sufficient criterion)**.** *To check that a finite non-empty subset H of G is a subgroup, it is enough to check that H is closed under the group operation.*

Example 2.5. (1) The multiples of 5 form a subgroup of $(\mathbb{Z}, +)$. The sum of two multiples of 5 is a multiple of 5, and so is the negative of a multiple of 5.

(2) The set

$$H = \left\{ \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} : a, b, d \in \mathbb{R},\ ad \neq 0 \right\}$$

is a subgroup of $\mathrm{GL}(2, \mathbb{R})$. Indeed $\left[\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right]\left[\begin{smallmatrix} a' & b' \\ 0 & d' \end{smallmatrix}\right] = \left[\begin{smallmatrix} aa' & ab'+bd' \\ 0 & dd' \end{smallmatrix}\right]$ and $aa'dd' \neq 0$. By the inverse formula of Example 1.7 the inverse of such a matrix has lower-left entry 0 again.

(3) The set $H = \{a + bi : a, b \in \mathbb{R},\ a^2 + b^2 = 1\}$ is a subgroup of $\mathbb{C}^*$ *under multiplication.* Here H is the set of complex numbers z with $|z| = 1$. For $z, w \in H$ we have $|zw| = |z||w| = 1$ and $|z^{-1}| = 1/|z| = 1$, and $1 \in H$, so the subgroup test applies. (Under *addition* H is not a subgroup, since $1 + 1 = 2 \notin H$.)

2.2 Congruence Modulo a Subgroup

Reading. Herstein, §2.4 (congruence modulo H); Gallian, Ch. 7 (properties of cosets).

Definition 2.6 (Congruence modulo H)**.** Let G be a group and H a subgroup of G. For $a, b \in G$ we say that a is *congruent to* b *modulo* H, written $a \equiv b \pmod{H}$, if $ab^{-1} \in H$.

Lemma 2.7. $a \equiv b \pmod{H}$ *is an equivalence relation on* G.

Proof. We must show, for all $a, b, c \in G$:

(1) *Reflexive:* $aa^{-1} = e \in H$, so $a \equiv a \pmod{H}$.

(2) *Symmetric:* If $a \equiv b \pmod{H}$ then $ab^{-1} \in H$. Since H is a subgroup, $(ab^{-1})^{-1} = ba^{-1} \in H$, so $b \equiv a \pmod{H}$.

(3) *Transitive:* If $ab^{-1} \in H$ and $bc^{-1} \in H$ then their product $(ab^{-1})(bc^{-1}) = ac^{-1} \in H$, so $a \equiv c \pmod{H}$. □

Remark 2.8. Let $G = \mathbb{Z}$ and $H = n\mathbb{Z}$, the multiples of n, with $\cdot$ equal to $+$. Then $a \equiv b \pmod{H}$ means $a - b \in H$, where $-b$ is the additive inverse of b. This is exactly the usual statement that a and b are equivalent modulo n.

2.3 Right Cosets

Reading. Gallian, Ch. 7 (Cosets and Lagrange's Theorem); Herstein, §2.4.

Definition 2.9 (Right coset)**.** Let H be a subgroup of G and let $a \in G$. The set

$$Ha = \{ ha : h \in H \}$$

is called the *right coset* of H in G through a.

For $a \in G$ write $[a] = \{x \in G : a \equiv x \pmod{H}\}$ for the equivalence class of a.

Lemma 2.10. *For every $a \in G$ we have $Ha = [a]$.*

Proof. We show $Ha \subseteq [a]$ and $[a] \subseteq Ha$.

First part: $Ha \subseteq [a]$. Let $h \in H$. Then

$$a(ha)^{-1} = a\,(a^{-1}h^{-1}) = h^{-1} \in H.$$

So $a \equiv ha \pmod{H}$, that is, $ha \in [a]$. This holds for every $h \in H$.

Second part: $[a] \subseteq Ha$. Let $x \in [a]$. Then $ax^{-1} \in H$. Since H is a subgroup, $(ax^{-1})^{-1} = xa^{-1} = h$ for some $h \in H$. Multiplying on the right by a gives $x = ha$, so $x \in Ha$. □

Remark 2.11. Lemma 2.10 says that every right coset is an equivalence class. Since $e \in H$, every element $g \in G$ lies in the coset Hg. So the right cosets cover G.

Lemma 2.12. *Any two right cosets of H in G are either identical or have no element in common.*

Proof. Let Ha and Hb be right cosets, and assume $Ha \cap Hb \neq \emptyset$. Let c be a common element, $c = h_1a = h_2b$ with $h_1, h_2 \in H$. We show $Ha = Hb$.

$Ha \subseteq Hb$. From $h_1a = h_2b$ we get $a = h_1^{-1}h_2b$. For any $h \in H$,

$$ha = hh_1^{-1}h_2\,b = (hh_1^{-1}h_2)\,b \in Hb,$$

because $hh_1^{-1}h_2 \in H$.

$Hb \subseteq Ha$. Symmetrically, $b = h_2^{-1}h_1a$ and for any $h \in H$, $hb = (hh_2^{-1}h_1)a \in Ha$. □

Lemma 2.13. *There is a one-to-one correspondence between any two right cosets of H in G.*

Proof. Let Ha and Hb be right cosets. Define $\varphi : Ha \to Hb$ by $\varphi(ha) = hb$.

- *Well-defined.* If $ha = h'a$ then $h = h'$ by cancellation, so $hb = h'b$.
- *Injective.* If $\varphi(h_1a) = \varphi(h_2a)$ then $h_1b = h_2b$, so $h_1 = h_2$ and $h_1a = h_2a$.
- *Surjective.* Every element of Hb has the form $hb = \varphi(ha)$ with $ha \in Ha$. □

Corollary 2.14. *Every right coset of H has exactly $|H|$ elements (take $a = e$, so $Ha = H$).*

2.4 Lagrange's Theorem

Reading. Gallian, Ch. 7 (Lagrange's theorem and consequences); Herstein, §2.4.

Definition 2.15 (Index)**.** If H is a subgroup of a finite group G, the *index* of H in G is the number of distinct right cosets of H in G. We write it $i_G(H)$.

Theorem 2.16 (Lagrange)**.** *If G is a finite group and H is a subgroup of G, then $|H|$ divides $|G|$. In fact*

$$i_G(H) = \frac{|G|}{|H|}.$$

Proof. By the remark after Lemma 2.10, the right cosets cover G. By Lemma 2.12, distinct right cosets are disjoint. So G is the disjoint union of its $k = i_G(H)$ distinct right cosets. By the corollary to Lemma 2.13, each coset has $|H|$ elements. Therefore $|G| = k\,|H|$, which gives $k = |G|/|H|$. □

A second proof, by listing elements. Let $H = \{h_1, \dots, h_r\}$. Choose $a_1 = e$ and write a first row $h_1a_1, \dots, h_ra_1$, which lists H. If some element of G is not yet listed, choose such an element a_2 and write a second row $h_1a_2, \dots, h_ra_2$. In general, if $a_1, \dots, a_t$ are chosen, and some element of G is not in $Ha_1 \cup \cdots \cup Ha_t$, choose such an element a_{t+1} and write the row $h_1a_{t+1}, \dots, h_ra_{t+1}$.

Claim: no element appears twice, in the same row or in different rows.

- Within a row: if $h_ia_t = h_ja_t$ then $h_i = h_j$ by cancellation.
- In rows $s < t$: if $h_ia_s = h_ja_t$ then $a_t = h_j^{-1}h_ia_s \in Ha_s$. This contradicts the choice of a_t outside $Ha_1 \cup \cdots \cup Ha_{t-1}$.

Each new row adds $r \geqslant 1$ new elements, and G is finite, so the process stops after some number k of rows, when all elements of G are listed. There is no repetition, so $|G| = k \cdot |H|$. □

Remark 2.17. The converse of Lagrange's theorem is false: if m divides $|G|$, then G need not have a subgroup of order m. The next proposition gives an example.

Proposition 2.18. *The alternating group A_4 (the group of even permutations of $\{1,2,3,4\}$) has order 12 but has no subgroup of order 6.*

Proof. The even permutations of $\{1,2,3,4\}$ are the identity, the three double transpositions $(12)(34), (13)(24), (14)(23)$, and the eight 3-cycles (choose the fixed point in 4 ways, and then one of the two 3-cycles on the other three points). So $|A_4| = 1 + 3 + 8 = 12$.

Suppose H is a subgroup of A_4 with $|H| = 6$. By Theorem 2.16 H has exactly two right cosets, and for $g \notin H$ these are H and Hg. We claim that $g^2 \in H$ for every $g \in A_4$. This is clear if $g \in H$. If $g \notin H$, then $g^2 \in H$ or $g^2 \in Hg$, because the two cosets cover A_4. If $g^2 = hg$ with $h \in H$, then $g = h \in H$, a contradiction. So $g^2 \in H$.

Now let c be a 3-cycle. Then $c^3 = e$, so $c = c^4 = (c^2)^2$, and $c^2 \in A_4$. Applying the claim to $g = c^2$ gives $c = g^2 \in H$. So H contains all eight 3-cycles, and $|H| \geqslant 8 > 6$, a contradiction. □

2.5 An Example in S_3

Reading. Gallian, Ch. 5 (Permutation Groups) and Ch. 7; Herstein, §2.10.

Let $S_3 = \{e, (12), (13), (23), (123), (132)\}$ and $H = \{e, (123), (132)\}$. We compose permutations from right to left: $\sigma\tau$ means "first τ, then σ". Then $\sigma \equiv \tau \pmod H$ means $\sigma\tau^{-1} \in H$.

Example 2.19. (1) $(12) \equiv (13) \pmod{H}$. Indeed $(12)(13)^{-1} = (12)(13)$. This sends $1 \mapsto 3$, $3 \mapsto 2$, $2 \mapsto 1$, which is $(132) \in H$.

(2) $(23) \equiv (12) \pmod{H}$. Indeed $(23)(12)^{-1} = (23)(12)$ sends $1 \mapsto 3$, $3 \mapsto 2$, $2 \mapsto 1$, which is $(132) \in H$.

(3) $e \equiv (123) \pmod{H}$. Indeed $e\,(123)^{-1} = (132) \in H$.
The equivalence classes (the right cosets) are

$$[e] = H, \qquad [(12)] = \{(12), (13), (23)\}.$$

There are exactly two right cosets, so $i_{S_3}(H) = 2 = 6/3$, as Lagrange's theorem predicts.

Exercises

E2.1 Prove that the intersection of any family of subgroups of G is a subgroup of G.

E2.2 Prove that $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.

E2.3 Let $|G| = p$ with p prime. Prove that G is cyclic and that its only subgroups are $\{e\}$ and G.

E2.4 Let G be finite and $a \in G$. Prove that $\operatorname{ord}(a)$ divides $|G|$, and that $a^{|G|} = e$.

E2.5 Deduce Fermat's little theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. (Use $U(p)$.)

E2.6 List all subgroups of S_3. For $H = \langle (1\,2) \rangle$ list the left cosets and the right cosets of H and compare them.

E2.7 Let H, K be finite subgroups of G with $\gcd(|H|, |K|) = 1$. Prove that $H \cap K = \{e\}$.

E2.8 Let $H \leqslant K \leqslant G$ with G finite. Prove that $i_G(H) = i_G(K)\, i_K(H)$.

More problems. Gallian: Ch. 3, Ch. 5 and Ch. 7, the exercise sets on subgroups, permutations, cosets and Lagrange's theorem. Herstein: §2.4 and §2.10, the problem sets.

3 Normal Subgroups and Quotient Groups

3.1 Left Cosets and Normal Subgroups

Reading. Gallian, Ch. 9 (Normal Subgroups and Factor Groups); Herstein, §2.6.

Definition 3.1 (Left coset)**.** Let H be a subgroup of G and $a \in G$. The set $aH = \{ah : h \in H\}$ is the *left coset* of H in G through a. (The right coset is $Ha = \{ha : h \in H\}$.)

Everything we proved for right cosets also holds for left cosets. Here is a short proof of the main facts.

Lemma 3.2. *Let H be a subgroup of G.*

(1) The left cosets of H cover G, and any two left cosets are equal or disjoint.

(2) Every left coset has $|H|$ elements.

(3) The number of left cosets of H equals the number of right cosets of H. For finite G it is $|G|/|H|$.

Proof. Let $\iota : G \to G$ be the map $\iota(x) = x^{-1}$. It is a bijection, because $\iota \circ \iota$ is the identity (Lemma 1.17 (3)). For $a \in G$,

$$\iota(Ha) = \{(ha)^{-1} : h \in H\} = \{a^{-1}h^{-1} : h \in H\} = a^{-1}H,$$

because h^{-1} runs over all of H when h does. So ι maps the right coset Ha onto the left coset $a^{-1}H$, and every left coset bH is the image of the right coset Hb^{-1}.

(1) The element $b \in G$ lies in bH, so the left cosets cover G. If $aH \cap bH \neq \emptyset$, apply ι (a bijection): then $Ha^{-1} \cap Hb^{-1} \neq \emptyset$, so $Ha^{-1} = Hb^{-1}$ by Lemma 2.12, and applying ι again gives $aH = bH$.

(2) $|aH| = |Ha^{-1}| = |H|$ because ι is a bijection and by the corollary to Lemma 2.13.

(3) The map $Ha \mapsto a^{-1}H$ is well-defined (if $Ha = Hb$ then applying ι gives $a^{-1}H = b^{-1}H$), and it is a bijection from the set of right cosets onto the set of left cosets, with inverse $bH \mapsto Hb^{-1}$. For finite G the count $|G|/|H|$ is Lagrange's theorem. □

Definition 3.3 (Normal subgroup)**.** A subgroup N of G is *normal* in G if $gng^{-1} \in N$ for every $g \in G$ and every $n \in N$.

Remark 3.4 (Observation). If N is normal, then for all $g \in G$ and $n \in N$ we have $gn = (gng^{-1})\, g \in Ng$ and $ng^{-1} = g^{-1}(gng^{-1}) \in g^{-1}N$.

For a subset $N \subseteq G$ and $g \in G$ write $gNg^{-1} = \{gng^{-1} : n \in N\}$.

Lemma 3.5. *A subgroup N of G is normal if and only if $gNg^{-1} = N$ for every $g \in G$.*

Proof. ($\Leftarrow$) If $gNg^{-1} = N$ then $gng^{-1} \in N$ for all $n \in N$, which is the definition.

($\Rightarrow$) Let N be normal and let $g \in G$. By the definition, $gNg^{-1} \subseteq N$. The same statement for the element g^{-1} gives $g^{-1}Ng \subseteq N$. Conjugating this inclusion by g gives

$$N = g\,(g^{-1}Ng)\,g^{-1} \;\subseteq\; g\,N\,g^{-1}.$$

Hence $gNg^{-1} = N$. □

Remark 3.6. Lemma 3.5 does *not* say that $gng^{-1} = n$ for each n. It says that the *set* gNg^{-1} equals the set N. The individual elements may be moved around inside N.

Lemma 3.7. *A subgroup N of G is normal if and only if every left coset of N in G is a right coset of N in G.*

Proof. ($\Rightarrow$) Let N be normal and $g \in G$. By Lemma 3.5, $gNg^{-1} = N$. Multiplying on the right by g gives $gN = Ng$. So the left coset gN is the right coset Ng.

($\Leftarrow$) Assume every left coset is a right coset. Let $g \in G$. Then $gN = Ng'$ for some $g' \in G$. Now $g \in gN$ (because $e \in N$), so $g \in Ng'$. Two right cosets that share the element g are equal, so $Ng' = Ng$. Hence $gN = Ng$ for every $g \in G$. Multiplying on the right by g^{-1} gives $gNg^{-1} = N$ for every g. By Lemma 3.5, N is normal. □

3.2 Products of Subgroups

Reading. Herstein, §2.5 (A Counting Principle); Gallian, Ch. 7 and Ch. 9 (exercises on HK).

Definition 3.8. If H, K are subgroups of G, let

$$HK = \{\, x \in G : x = hk,\ h \in H,\ k \in K \,\}.$$

Lemma 3.9. *HK is a subgroup of G if and only if $HK = KH$.*

Proof. ($\Leftarrow$) Assume $HK = KH$. The set HK contains $e = ee$, and associativity is inherited from G.

- *Closure.* Take $hk,\ h'k' \in HK$. Then $(hk)(h'k') = h\,(kh')\,k'$. Now $kh' \in KH = HK$, so $kh' = h''k''$ for some $h'' \in H$, $k'' \in K$. Hence $(hk)(h'k') = (hh'')(k''k') \in HK$.
- *Inverses.* $(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$.

($\Rightarrow$) Assume HK is a subgroup. Let $hk \in HK$. Then $(hk)^{-1} \in HK$, say $(hk)^{-1} = h'k'$. So $hk = (h'k')^{-1} = k'^{-1}h'^{-1} \in KH$. This shows $HK \subseteq KH$. Now let $kh \in KH$. Then $h^{-1}k^{-1} \in HK$, and since HK is a subgroup, its inverse $(h^{-1}k^{-1})^{-1} = kh$ lies in HK. This shows $KH \subseteq HK$. □

Theorem 3.10. *If H and K are finite subgroups of G, then*

$$|HK| = \frac{|H|\,|K|}{|H \cap K|}.$$

Proof. Consider the map $\pi : H \times K \to HK$, $\pi(h, k) = hk$. It is onto by definition of HK. We count the number of pairs that map to a given element $x = hk \in HK$.

Suppose $h'k' = hk$ with $h' \in H$, $k' \in K$. Put $t = h^{-1}h'$. Then $t \in H$, and also $t = kk'^{-1} \in K$ (from $h^{-1}h' = kk'^{-1}$). So $t \in H \cap K$, and $h' = ht$, $k' = t^{-1}k$. Conversely, for every $t \in H \cap K$ the pair $(ht,\ t^{-1}k)$ lies in $H \times K$ and maps to $ht\,t^{-1}k = hk$. Different t give different pairs. So every element of HK has exactly $|H \cap K|$ preimages. Therefore $|H|\,|K| = |HK| \cdot |H \cap K|$. □

Remark 3.11. In Theorem 3.10 the set HK need not be a subgroup. The theorem counts the elements of the set HK.

Lemma 3.12. *For every subgroup H of G we have $HH = H$.*

Proof. $HH \subseteq H$ by closure. For $h \in H$ we have $h = he \in HH$, so $H \subseteq HH$. □

3.3 Quotient Groups

Reading. Gallian, Ch. 9 (Factor Groups); Herstein, §2.6.

Lemma 3.13. *Let N be a normal subgroup of G and $a, b \in G$. Then $(Na)(Nb) = Nab$.*

Proof. By Lemma 3.7, $aN = Na$. So, using associativity of set products and Lemma 3.12,

$$NaNb = N(aN)b = N(Na)b = (NN)ab = Nab.$$ □

Lemma 3.14. *A subgroup N of G is normal if and only if the product of two right cosets of N in G is again a right coset of N in G.*

Proof. ($\Rightarrow$) This is Lemma 3.13.

($\Leftarrow$) Assume that the product of any two right cosets is a right coset. Let $g \in G$. Consider $(Ng)(Ng^{-1})$. By assumption it equals Nc for some $c \in G$. Since $g \in Ng$ and $g^{-1} \in Ng^{-1}$, we have $e = gg^{-1} \in (Ng)(Ng^{-1}) = Nc$. Also $e \in N$. The cosets Nc and N share the element e, so $Nc = N$. Hence

$$(Ng)(Ng^{-1}) = N.$$

Now let $n, n' \in N$. Then $(ng)(n'g^{-1}) \in N$, say $ng\,n'g^{-1} = n''$ with $n'' \in N$. Therefore $gn'g^{-1} = n^{-1}n'' \in N$. Since $g \in G$ and $n' \in N$ were arbitrary, N is normal. □

Definition 3.15 (Quotient group)**.** Let N be a normal subgroup of G. The collection of all right cosets of N in G is written G/N.

Theorem 3.16. *If N is a normal subgroup of G, then G/N is a group under the operation $(Na)(Nb) = Nab$.*

Proof. By Lemma 3.13 the operation is well-defined on cosets (it is the set product) and G/N is closed under it. Further:

- *Associativity:* $\big((Na)(Nb)\big)(Nc) = N(abc) = (Na)\big((Nb)(Nc)\big)$.
- *Identity:* $Ne = N$ satisfies $(Ne)(Na) = Na = (Na)(Ne)$.
- *Inverses:* $(Na)(Na^{-1}) = N = (Na^{-1})(Na)$. □

Remark 3.17. Take $G = \mathbb{Z}$ and $N = n\mathbb{Z}$. The cosets are $a + n\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is the group of integers modulo n.

Exercises

E3.1 Prove that a subgroup of index 2 is normal.

E3.2 The *centre* of G is $Z(G) = \{z \in G : zg = gz \ \forall g \in G\}$. Prove that $Z(G)$ is a normal subgroup of G.

E3.3 Let N be normal in G and H a subgroup of G. Prove that $NH = HN$ and that NH is a subgroup of G.

E3.4 Show that $A_3 = \{e, (1\,2\,3), (1\,3\,2)\}$ is normal in S_3 but $\langle (1\,2) \rangle$ is not.

E3.5 Prove that if $G/Z(G)$ is cyclic, then G is abelian.

E3.6 For $a, b \in G$ put $[a, b] = aba^{-1}b^{-1}$. Let N be normal in G. Prove that G/N is abelian if and only if $[a, b] \in N$ for all $a, b \in G$.

E3.7 Let N be normal in G and $a \in G$ of finite order. Prove that the order of Na in G/N divides the order of a.

E3.8 Let H, K be subgroups of G with K normal. Prove that $H \cap K$ is normal in H.

More problems. Gallian: Ch. 9, the exercise set on normal subgroups and factor groups; Ch. 7 for $|HK|$. Herstein: §2.5 and §2.6, the problem sets.

4 Group Homomorphisms

4.1 Definition and Examples

Reading. Gallian, Ch. 10 (Group Homomorphisms); Herstein, §2.7; Gallian, Ch. 6 for isomorphisms.

Definition 4.1 (Homomorphism)**.** A map $\varphi : G \to G'$ is a *homomorphism* if

$$\varphi(ab) = \varphi(a)\varphi(b) \qquad \text{for all } a, b \in G.$$

On the left ab is computed with the operation of G. On the right $\varphi(a)\varphi(b)$ is computed with the operation of G'. So a homomorphism turns the operation of G into the operation of G'. The operations do not have to look like multiplication. For example, a homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$ satisfies $\varphi(x + y) = \varphi(x)\varphi(y)$.

Example 4.2. (1) The identity map $\varphi(x) = x$ from G to G. The *trivial homomorphism* $\varphi(x) = e'$ for all $x \in G$ is also a homomorphism, since $e'e' = e'$.

(2) $G = (\mathbb{R}, +)$, $G' = (\mathbb{R}_{>0}, \times)$, $\varphi(x) = 2^x$. Indeed $2^{x+y} = 2^x 2^y$. (We take $\mathbb{R}_{>0}$ and not $\mathbb{R}$ as the target, because $\mathbb{R}$ is not a group under multiplication: 0 has no inverse.)

(3) $G = \mathbb{Z}_n = \{0, \dots, n-1\}$ under addition modulo n, and $G' = \{e^{2\pi ik/n} : 0 \leqslant k < n\}$, the n-th roots of unity, under multiplication. (G' is a group: $e^{2\pi ik/n} e^{2\pi il/n} = e^{2\pi i(k+l)/n}$ is again an n-th root of unity because $e^{2\pi i} = 1$, and the inverse of $e^{2\pi ik/n}$ is $e^{2\pi i(n-k)/n}$.) Define $\varphi(k) = e^{2\pi ik/n}$. If $k + l < n$ then $\varphi(k + l) = \varphi(k)\varphi(l)$ directly. If $k + l \geqslant n$ the sum modulo n is $k + l - n$, and $e^{2\pi i(k+l-n)/n} = e^{2\pi i(k+l)/n}$ because $e^{2\pi i} = 1$. So $\varphi(k \oplus l) = \varphi(k)\varphi(l)$.

(4) The determinant $\det : \mathrm{GL}(2, \mathbb{R}) \to \mathbb{R}^*$, because $\det(AB) = \det A \det B$ (computed in Example 1.7).

(5) The absolute value $x \mapsto |x|$ from $\mathbb{R}^*$ to $\mathbb{R}_{>0}$, because $|xy| = |x||y|$. The sign map $s(x) = x/|x|$ from $\mathbb{R}^*$ to $\{1, -1\}$ is also a homomorphism, because $s(xy) = \dfrac{xy}{|x||y|} = s(x)s(y)$.

(6) Reduction modulo d. Let d divide m. The map $\mathbb{Z}_m \to \mathbb{Z}_d$, $a \bmod m \mapsto a \bmod d$, is a homomorphism of additive groups. It is well-defined because if $a \equiv b \pmod m$ then $m \mid a - b$, so $d \mid a - b$.

(7) The sign $\operatorname{sgn} : S_n \to \{1, -1\}$ of a permutation is a homomorphism (Proposition 4.3 below).

(8) *Conjugation.* Fix $g \in G$ and define $\gamma_g : G \to G$ by $\gamma_g(x) = gxg^{-1}$. Then

$$\gamma_g(x)\gamma_g(y) = gxg^{-1}gyg^{-1} = gxyg^{-1} = \gamma_g(xy).$$

So γ_g is a homomorphism.

Proposition 4.3 (The sign of a permutation)**.** *For $\sigma \in S_n$ define*

$$\operatorname{sgn}(\sigma) = \prod_{1 \leqslant i < j \leqslant n} \frac{\sigma(j) - \sigma(i)}{j - i}.$$

Then $\operatorname{sgn}(\sigma) \in \{1, -1\}$ and $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$ for all $\sigma, \tau \in S_n$. So $\operatorname{sgn} : S_n \to \{1, -1\}$ is a homomorphism.

Proof. Let P be the set of unordered pairs $\{i, j\}$ with $i \neq j$ in $\{1, \dots, n\}$. For $\sigma \in S_n$ put $f_\sigma(\{i, j\}) = \dfrac{\sigma(j) - \sigma(i)}{j - i}$. Exchanging i and j changes the sign of both the numerator and the denominator, so f_σ is well-defined on unordered pairs, and $\operatorname{sgn}(\sigma) = \prod_{p \in P} f_\sigma(p)$.

The map $\{i, j\} \mapsto \{\sigma(i), \sigma(j)\}$ is a bijection of P (its inverse uses σ^{-1}). Hence the products $\prod_P |\sigma(j) - \sigma(i)|$ and $\prod_P |j - i|$ are equal, since they have the same factors in a different order. So $|\operatorname{sgn}(\sigma)| = 1$. As $\operatorname{sgn}(\sigma)$ is a real number, $\operatorname{sgn}(\sigma) = \pm 1$.

For the product rule, for every pair $\{i, j\}$,

$$f_{\sigma\tau}(\{i, j\}) = \frac{\sigma(\tau(j)) - \sigma(\tau(i))}{\tau(j) - \tau(i)} \cdot \frac{\tau(j) - \tau(i)}{j - i} = f_\sigma(\{\tau(i), \tau(j)\})\, f_\tau(\{i, j\}).$$

Take the product over all $\{i, j\} \in P$. Since $\{i, j\} \mapsto \{\tau(i), \tau(j)\}$ is a bijection of P, $\prod_P f_\sigma(\{\tau(i), \tau(j)\}) = \prod_P f_\sigma(p) = \operatorname{sgn}(\sigma)$. So $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$. □

The kernel of sgn is the set of permutations with sign $+1$; they are called *even* permutations, and this group is the alternating group A_n.

Non-example 4.4. (1) $\varphi : \mathbb{Z} \to \mathbb{Z}$, $\varphi(x) = x + 1$, is not a homomorphism: $\varphi(0+0) = 1$, but $\varphi(0) + \varphi(0) = 2$. (We will see in Theorem 4.5 that $\varphi(e)$ must equal e'.)

(2) Squaring is a homomorphism $G \to G$ exactly when G is abelian, because $(xy)^2 = x^2y^2 \iff xyxy = xxyy \iff yx = xy$. For example, squaring on $\mathrm{GL}(2,\mathbb{R})$ is not a homomorphism: take $A = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$ and $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$. Then $AB = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ and $(AB)^2 = \left[\begin{smallmatrix} 5 & 3 \\ 3 & 2 \end{smallmatrix}\right]$, but $A^2B^2 = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 5 & 2 \\ 2 & 1 \end{smallmatrix}\right]$.

4.2 Basic Properties

Reading. Gallian, Ch. 10 (properties of homomorphisms); Herstein, §2.7.

Theorem 4.5. *Let $\varphi : G \to G'$ be a homomorphism. Then*

(1) $\varphi(e) = e'$;

(2) $\varphi(x^{-1}) = \varphi(x)^{-1}$ *for all* $x \in G$;

(3) $\varphi(x^n) = \varphi(x)^n$ *for all* $x \in G$ *and all* $n \in \mathbb{Z}$.

Proof. (1) For any $x \in G$,

$$\varphi(x)\, e' = \varphi(x) = \varphi(xe) = \varphi(x)\varphi(e).$$

Cancelling $\varphi(x)$ on the left (Lemma 1.17 (0)) gives $e' = \varphi(e)$.

(2) We have $e' = \varphi(e) = \varphi(xx^{-1}) = \varphi(x)\varphi(x^{-1})$. So $\varphi(x^{-1})$ is an inverse of $\varphi(x)$, and by uniqueness of inverses $\varphi(x^{-1}) = \varphi(x)^{-1}$.

(3) For $n = 0$ this is (1). For $n \geqslant 1$ use induction: the case $n = 1$ is trivial, and if $\varphi(x^n) = \varphi(x)^n$ then $\varphi(x^{n+1}) = \varphi(x^n)\varphi(x) = \varphi(x)^{n+1}$. For $n = -N < 0$,

$$\varphi(x^{-N}) = \varphi\big((x^N)^{-1}\big) = \varphi(x^N)^{-1} = \big(\varphi(x)^N\big)^{-1} = \varphi(x)^{-N}. \qquad \square$$

Corollary 4.6. *If $\varphi : G \to G'$ is a homomorphism and $x^n = e$, then $\varphi(x)^n = e'$. In particular, if x has order n, then the order of $\varphi(x)$ divides n.*

Proof. $\varphi(x)^n = \varphi(x^n) = \varphi(e) = e'$. The order of an element divides every positive integer that sends it to the identity. $\square$

Theorem 4.7 (Composition)**.** *If $\varphi_1 : G_1 \to G_2$ and $\varphi_2 : G_2 \to G_3$ are homomorphisms, then $\varphi_2 \circ \varphi_1 : G_1 \to G_3$ is a homomorphism.*

Proof. For $x, y \in G_1$: $(\varphi_2 \circ \varphi_1)(xy) = \varphi_2\big(\varphi_1(x)\varphi_1(y)\big) = \varphi_2(\varphi_1(x))\,\varphi_2(\varphi_1(y))$. $\square$

Theorem 4.8 (Images and inverse images of subgroups)**.** *Let $\varphi : G \to G'$ be a homomorphism.*

(1) If K is a subgroup of G, then $\varphi(K) = \{\varphi(k) : k \in K\}$ is a subgroup of G'. In particular $\varphi(G) = \operatorname{Im}\varphi$ is a subgroup of G'.

(2) If L is a subgroup of G', then $\varphi^{-1}(L) = \{x \in G : \varphi(x) \in L\}$ is a subgroup of G.

Proof. We use the subgroup test (Lemma 2.2).

(1) $\varphi(K) \ni \varphi(e)$ is non-empty. Two elements of $\varphi(K)$ have the form $\varphi(k)$ and $\varphi(k')$ with $k, k' \in K$. Then $\varphi(k)\varphi(k') = \varphi(kk') \in \varphi(K)$, and $\varphi(k)^{-1} = \varphi(k^{-1}) \in \varphi(K)$.

(2) $e \in \varphi^{-1}(L)$ because $\varphi(e) = e' \in L$. If $x, y \in \varphi^{-1}(L)$, then $\varphi(xy) = \varphi(x)\varphi(y) \in L$, so $xy \in \varphi^{-1}(L)$. Also $\varphi(x^{-1}) = \varphi(x)^{-1} \in L$, so $x^{-1} \in \varphi^{-1}(L)$. $\square$

4.3 The Kernel

Reading. Gallian, Ch. 10 (properties of kernels); Herstein, §2.7.

Definition 4.9 (Kernel)**.** The *kernel* of a homomorphism $\varphi : G \to G'$ is

$$\operatorname{Ker}\varphi = K_\varphi = \{\, x \in G : \varphi(x) = e' \,\}.$$

Theorem 4.10. *The kernel of a homomorphism $\varphi : G \to G'$ is a normal subgroup of G.*

Proof. $K_\varphi = \varphi^{-1}(\{e'\})$ is the inverse image of the trivial subgroup, so it is a subgroup of G by Theorem 4.8. For normality, let $g \in G$ and $k \in K_\varphi$. Then

$$\varphi(gkg^{-1}) = \varphi(g)\, e'\, \varphi(g^{-1}) = \varphi(g)\varphi(g)^{-1} = e'.$$

So $gkg^{-1} \in K_\varphi$ for all $g \in G$ and $k \in K_\varphi$. □

Example 4.11. (1) The kernel of $\det : \mathrm{GL}(2,\mathbb{R}) \to \mathbb{R}^*$ is the set of matrices of determinant 1.

(2) The kernel of the squaring map $\mathbb{R}^* \to \mathbb{R}^*$ is $\{1, -1\}$.

(3) The kernel of $\operatorname{sgn} : S_n \to \{\pm 1\}$ is the set of even permutations.

Theorem 4.12 (Injectivity)**.** *A homomorphism $\varphi : G \to G'$ is injective if and only if* $\operatorname{Ker}\varphi = \{e\}$.

Proof. ($\Rightarrow$) If $\varphi(x) = e' = \varphi(e)$, then $x = e$ by injectivity.

($\Leftarrow$) Suppose $\operatorname{Ker}\varphi = \{e\}$ and $\varphi(x) = \varphi(y)$. Then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = e'$, so $xy^{-1} \in \operatorname{Ker}\varphi = \{e\}$. Hence $x = y$. □

Theorem 4.13 (Fibres of a homomorphism)**.** *Let $\varphi : G \to G'$ be a homomorphism with kernel K. For $x, y \in G$,*

$$\varphi(x) = \varphi(y) \iff x^{-1}y \in K \iff y \in xK.$$

Since K is normal, $xK = Kx$. So the sets on which φ takes a single value are exactly the cosets of K. If G is finite, every value in $\operatorname{Im}\varphi$ *is taken exactly $|K|$ times.*

Proof. $\varphi(x^{-1}y) = \varphi(x)^{-1}\varphi(y)$, and this equals e' exactly when $\varphi(x) = \varphi(y)$. This proves the first equivalence, and the second is the definition of xK. The coset xK has $|K|$ elements by Lemma 2.13 (applied to left cosets, see Lemma 3.2). □

4.4 Every Normal Subgroup Is a Kernel

Reading. Gallian, Ch. 10 and Ch. 9 (natural map); Herstein, §2.7 and §2.8 (inner automorphisms).

Lemma 4.14. *Let N be a normal subgroup of G. Define $\pi : G \to G/N$ by $\pi(x) = Nx$. Then π is a homomorphism from G onto G/N, and* $\operatorname{Ker}\pi = N$.

Proof. By Lemma 3.13, $\pi(xy) = Nxy = (Nx)(Ny) = \pi(x)\pi(y)$. Every coset Nx is $\pi(x)$, so π is onto. The identity of G/N is N, and $\pi(x) = N$ if and only if $Nx = N$, that is, if and only if $x \in N$. So $\operatorname{Ker}\pi = N$. □

Theorem 4.10 and Lemma 4.14 together show: *a subgroup of G is normal if and only if it is the kernel of some homomorphism defined on G.*

Exercises

E4.1 Let $\varphi : G \to H$ be a homomorphism. Prove that $\varphi(e) = e$ and $\varphi(a^{-1}) = \varphi(a)^{-1}$.

E4.2 Prove that φ is injective if and only if $\operatorname{Ker}\varphi = \{e\}$.

E4.3 Prove that $\det : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^*$ is a homomorphism, and find its kernel and image.

E4.4 Prove that the image of a homomorphism is a subgroup of the target. Prove that if G is abelian, so is $\varphi(G)$.

E4.5 Prove that $\operatorname{ord}(\varphi(a))$ divides $\operatorname{ord}(a)$ whenever a has finite order.

E4.6 Find all homomorphisms $\mathbb{Z} \to \mathbb{Z}$. Find all homomorphisms $\mathbb{Z}_{12} \to \mathbb{Z}_5$.

E4.7 For $g \in G$ define $c_g : G \to G$ by $c_g(x) = gxg^{-1}$. Prove that c_g is a bijective homomorphism.

E4.8 Prove that $\operatorname{sgn} : S_n \to \{\pm 1\}$ is onto for $n \geqslant 2$, and that its kernel A_n has index 2.

More problems. Gallian: Ch. 6 and Ch. 10, the exercise sets on isomorphisms and homomorphisms. Herstein: §2.7 and §2.8, the problem sets.

5 Rings, Integral Domains, and Fields

5.1 Rings

Reading. Gallian, Ch. 12 (Introduction to Rings); Herstein, §3.1.

Definition 5.1 (Associative ring)**.** A non-empty set R is an *associative ring* if two operations, written $+$ and $\cdot$, are defined on R such that for all $a, b, c \in R$:

(R1) $a + b \in R$.

(R2) $a + b = b + a$.

(R3) $(a + b) + c = a + (b + c)$.

(R4) There is an element $0 \in R$ such that $a + 0 = a$ for every $a \in R$.

(R5) For every $a \in R$ there is an element $-a \in R$ such that $a + (-a) = 0$.

(R6) $a \cdot b \in R$.

(R7) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.

(R8) $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(b + c) \cdot a = b \cdot a + c \cdot a$ (the *distributive laws*).

Remark 5.2. (1) (R1)–(R5) say that R is an *abelian group under* $+$.

(2) A *non-associative ring* is a ring in which associativity of $\cdot$ fails.

(3) A ring is *commutative* if $a \cdot b = b \cdot a$ for all $a, b \in R$.

(4) We shall refer to $+$ as *addition* and to $\cdot$ (or $\times$) as *multiplication*.

Definition 5.3 (Ring with unit element)**.** If there is an element $1 \in R$ such that $1 \cdot a = a \cdot 1 = a$ for every $a \in R$, then R is a *ring with unit element* (or ring with unity).

A ring may or may not contain 1. For example, the ring of even integers $2\mathbb{Z}$ (with the usual $+$ and $\times$) does not contain 1: a unit element $u \in 2\mathbb{Z}$ would satisfy $u \cdot 2 = 2$, so $u = 1$, which is odd.

Example 5.4. (1) $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ with the usual addition and multiplication. All eight ring axioms are the familiar rules of arithmetic.

(2) $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ with addition and multiplication modulo n; for example $\mathbb{Z}_7$. Write $a \oplus b$ and $a \odot b$ for the remainders of $a + b$ and ab modulo n. If $a \equiv a'$ and $b \equiv b' \pmod n$ then $a + b \equiv a' + b'$ and $ab \equiv a'b'$. So each side of each ring axiom is a number in $\{0, \dots, n-1\}$ that is congruent modulo n to the corresponding integer expression (for example both $a \odot (b \oplus c)$ and $(a \odot b) \oplus (a \odot c)$ are congruent to $ab + ac$). Two numbers in $\{0, \dots, n-1\}$ that are congruent modulo n are equal, so every axiom holds because it holds in $\mathbb{Z}$. The ring $\mathbb{Z}_n$ is commutative with unity 1. (For prime $n = p$, the non-zero elements form an abelian group under multiplication; see Corollary 5.13.)

(3) *Pairs of reals.* On the set of pairs (α, β) with $\alpha, \beta \in \mathbb{R}$ define

$$(\alpha, \beta) + (\gamma, \delta) = (\alpha + \gamma,\ \beta + \delta), \qquad (\alpha, \beta) \cdot (\gamma, \delta) = (\alpha\gamma - \beta\delta,\ \alpha\delta + \beta\gamma).$$

Identify (α, β) with the complex number $\alpha + \beta i$. Then these operations are exactly addition and multiplication of complex numbers, since $(\alpha + \beta i)(\gamma + \delta i) = (\alpha\gamma - \beta\delta) + (\alpha\delta + \beta\gamma)i$. So this is the ring $\mathbb{C}$ written with pairs, and the ring axioms hold because they hold in $\mathbb{C}$.

(4) *Real quaternions.* The elements are $\alpha_0 + \alpha_1 i + \alpha_2 j + \alpha_3 k$ with $\alpha_t \in \mathbb{R}$. Addition is coordinate-wise. Multiplication is determined by

$$i^2 = j^2 = k^2 = ijk = -1, \qquad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j.$$

This is a *non-commutative* ring with $0 = 0 + 0i + 0j + 0k$ and $1 = 1 + 0i + 0j + 0k$ (Proposition 5.5 below).

Proposition 5.5 (The real quaternions form a ring)**.** *Let $\iota = \sqrt{-1} \in \mathbb{C}$. Let Q be the set of 2×2 complex matrices $\left[\begin{smallmatrix} a & b \\ -\bar{b} & \bar{a} \end{smallmatrix}\right]$ with $a, b \in \mathbb{C}$. Then Q is a non-commutative ring under matrix addition and multiplication with unity I. Put*

$$\mathbf{1} = I, \quad \mathbf{i} = \begin{bmatrix} \iota & 0 \\ 0 & -\iota \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 & \iota \\ \iota & 0 \end{bmatrix}.$$

Then these satisfy the relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -I$, $\mathbf{ij} = -\mathbf{ji} = \mathbf{k}$, $\mathbf{jk} = -\mathbf{kj} = \mathbf{i}$, $\mathbf{ki} = -\mathbf{ik} = \mathbf{j}$, and every element of Q can be written in exactly one way as $\alpha_0\mathbf{1} + \alpha_1\mathbf{i} + \alpha_2\mathbf{j} + \alpha_3\mathbf{k}$ with $\alpha_t \in \mathbb{R}$. So Q is the ring of real quaternions.

Proof. *Q is a subring of $M_2(\mathbb{C})$.* The 2×2 complex matrices form a ring under matrix addition and multiplication (associativity and distributivity hold for matrices). It remains to check that Q contains 0 and is closed under addition, negatives and multiplication; then all ring axioms are inherited. Closure under addition and negatives is clear. For multiplication,

$$\begin{bmatrix} a & b \\ -\bar{b} & \bar{a} \end{bmatrix} \begin{bmatrix} c & d \\ -\bar{d} & \bar{c} \end{bmatrix} = \begin{bmatrix} ac - b\bar{d} & ad + b\bar{c} \\ -\bar{b}c - \bar{a}\bar{d} & -\bar{b}d + \bar{a}\bar{c} \end{bmatrix}.$$

Put $u = ac - b\bar{d}$ and $v = ad + b\bar{c}$. Then $\bar{u} = \bar{a}\bar{c} - \bar{b}d$ and $-\bar{v} = -\bar{a}\bar{d} - \bar{b}c$. So the product equals $\left[\begin{smallmatrix} u & v \\ -\bar{v} & \bar{u} \end{smallmatrix}\right] \in Q$. Also $I \in Q$ (take $a = 1, b = 0$).

Relations. Direct matrix multiplication gives $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -\mathrm{I}$, $\mathbf{ij} = \mathbf{k} = -\mathbf{ji}$, $\mathbf{jk} = \mathbf{i} = -\mathbf{kj}$ and $\mathbf{ki} = \mathbf{j} = -\mathbf{ik}$. For example $\mathbf{ij} = \left[\begin{smallmatrix} \iota & 0 \\ 0 & -\iota \end{smallmatrix}\right]\left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & \iota \\ \iota & 0 \end{smallmatrix}\right] = \mathbf{k}$. Then $\mathbf{ijk} = \mathbf{kk} = -\mathrm{I}$. Since $\mathbf{ij} = \mathbf{k} \neq -\mathbf{k} = \mathbf{ji}$, the ring Q is not commutative.

Unique form. We have

$$\alpha_0\mathbf{1} + \alpha_1\mathbf{i} + \alpha_2\mathbf{j} + \alpha_3\mathbf{k} = \begin{bmatrix} \alpha_0 + \alpha_1\iota & \alpha_2 + \alpha_3\iota \\ -\alpha_2 + \alpha_3\iota & \alpha_0 - \alpha_1\iota \end{bmatrix},$$

which is the element of Q with $a = \alpha_0 + \alpha_1\iota$ and $b = \alpha_2 + \alpha_3\iota$ (note that $-\bar{b} = -\alpha_2 + \alpha_3\iota$ and $\bar{a} = \alpha_0 - \alpha_1\iota$). Every pair (a, b) of complex numbers has exactly one such real decomposition, so every element of Q has exactly one such expression. □

5.2 Zero Divisors, Integral Domains, and Fields

Reading. Gallian, Ch. 13 (Integral Domains); Herstein, §3.2.

Definition 5.6. Let R be a commutative ring.

(1) A non-zero element $a \in R$ is a *zero divisor* if there is $b \in R$, $b \neq 0$, with $ab = 0$.

(2) An *integral domain* is a commutative ring with unity $1 \neq 0$ and no zero divisors.

(3) A ring R is a *division ring* if its non-zero elements form a group under multiplication.

(4) A *field* is a commutative division ring.

Definition 5.7 (Unit)**.** An element a of a ring with unity is a *unit* if a^{-1} exists, that is, there is b with $ab = ba = 1$. Thus a field is a commutative ring with unity in which every non-zero element is a unit.

Remark 5.8. Some books define an integral domain without requiring a unit element. We always require a unit element $1 \neq 0$, as in Definition 5.6.

Lemma 5.9. *Let* R *be a ring and* $a, b \in R$. *Then*

(1) $a \cdot 0 = 0 \cdot a = 0$*;*

(2) $a(-b) = (-a)b = -(ab)$*;*

(3) $(-a)(-b) = ab$.

If R *has a unit element, then also*

(4) $(-1)a = -a$*;*

(5) $(-1)(-1) = 1$.

Proof. (1) $a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0$. Since $(R, +)$ is a group, we can cancel $a \cdot 0$ to get $a \cdot 0 = 0$. Similarly $0 \cdot a = 0$.

(2) $a(-b) + ab = a(-b + b) = a \cdot 0 = 0$. So $a(-b)$ is the additive inverse of ab, that is, $a(-b) = -(ab)$. The proof of $(-a)b = -(ab)$ is the same.

(3) By (2) applied twice, $(-a)(-b) = -\big(a(-b)\big) = -\big(-(ab)\big) = ab$.

(4) By (2), $(-1)a = -(1 \cdot a) = -a$.

(5) By (3), $(-1)(-1) = 1 \cdot 1 = 1$. □

Theorem 5.10 (Uniqueness)**.** *If a ring has a unit element, it is unique. If an element of a ring with unity has a multiplicative inverse, the inverse is unique.*

Proof. Let 1 and $1'$ be two unit elements. Then $1 = 1 \cdot 1'$ (as $1'$ is a unit element) and $1 \cdot 1' = 1'$ (as 1 is a unit element). So $1 = 1'$.

Let b and c both be inverses of a: $ab = ba = 1$ and $ac = ca = 1$. Then

$$b = b \cdot 1 = b(ac) = (ba)c = 1 \cdot c = c.$$

We did not need cancellation here; we used only associativity and the two equations. □

Remark 5.11. In a group, $xa = ya$ implies $x = y$. In a ring this can fail. We only get $(x - y)a = 0$. This does not force $x = y$ when a is a zero divisor.

Theorem 5.12. *A finite integral domain is a field.*

First proof. Let D be a finite integral domain and let $a \in D$, $a \neq 0$. Define $f : D \to D$ by $f(x) = xa$. This map is injective: if $xa = ya$ then $(x - y)a = 0$, and since $a \neq 0$ and D has no zero divisors, $x - y = 0$. An injective map from a finite set to itself is onto. So there is $b \in D$ with $ba = f(b) = 1$. Since D is commutative, $ab = ba = 1$ and a is a unit. So every non-zero element is a unit, and D is a field. □

Second proof. Let $a \neq 0$ in D. The powers $a, a^2, a^3, \ldots$ lie in D, which is finite. So there are $i < j$ with $a^i = a^j$. Then $a^i(a^{j-i} - 1) = 0$. Since $a^i \neq 0$ (a product of non-zero elements is non-zero in a domain) and D has no zero divisors, $a^{j-i} = 1$. So $a \cdot a^{j-i-1} = 1$, where $a^0 = 1$. Hence a^{j-i-1} is the inverse of a. □

Corollary 5.13. *If p is prime, then $\mathbb{Z}_p$ is a field.*

Proof. $\mathbb{Z}_p$ is a finite commutative ring with unity $1 \neq 0$. It has no zero divisors: if $ab \equiv 0 \pmod{p}$ then $p \mid ab$, so $p \mid a$ or $p \mid b$ because p is prime. So $\mathbb{Z}_p$ is a finite integral domain, hence a field by Theorem 5.12. □

5.3 The Characteristic of a Ring

Reading. Gallian, Ch. 13 (characteristic of a ring); Herstein, §3.2.

Definition 5.14 (Characteristic)**.** Let R be a ring with unity. If 1 has infinite order under addition, the *characteristic* of R is 0. If 1 has order n under addition (that is, n is the least positive integer with $\underbrace{1 + 1 + \cdots + 1}_{n} = 0$), the characteristic of R is n.

Theorem 5.15. *The characteristic of an integral domain is either 0 or a prime number.*

Proof. Suppose the characteristic is $n > 0$. Since $1 \neq 0$ in an integral domain, $n \geqslant 2$. Assume, for contradiction, that $n = st$ with $1 < s, t < n$. Then

$$0 = n \cdot 1 = (st) \cdot 1 = (s \cdot 1)(t \cdot 1)$$

by the distributive laws. Since there are no zero divisors, $s \cdot 1 = 0$ or $t \cdot 1 = 0$. But n is the least positive integer with $n \cdot 1 = 0$, and $s, t < n$. This is a contradiction. So n has no such factorisation, and n is prime. □

Example 5.16 (Roots of a polynomial)**.** Consider $x^2 - 4x + 3 = 0$. Over $\mathbb{Z}$ we can write $x^2 - 4x + 3 = (x-3)(x-1) = 0$. Since $\mathbb{Z}$ has no zero divisors, $x = 3$ or $x = 1$: there are two roots.

Over $\mathbb{Z}_{12}$ the answer is different. We compute $f(x) = x^2 - 4x + 3 = (x-1)(x-3)$ modulo 12 for all twelve residues:

x	0	1	2	3	4	5	6	7	8	9	10	11
$f(x) \bmod 12$	3	0	11	0	3	8	3	0	11	0	3	8

So $f(x) \equiv 0$ for exactly four values, $x = 1, 3, 7, 9$. For example $7^2 - 4 \cdot 7 + 3 = 24 \equiv 0 \pmod{12}$. The reason is that $\mathbb{Z}_{12}$ has zero divisors (for instance $3 \cdot 4 = 0$), so $(x-1)(x-3) = 0$ no longer forces one factor to be 0. Over $\mathbb{Z}_{11}$ and $\mathbb{Z}_{13}$, which are fields by Corollary 5.13, a product is zero only if a factor is zero, so $(x-1)(x-3) = 0$ has only the two roots $x = 1$ and $x = 3$.

Exercises

E5.1 In any ring R prove that $0a = a0 = 0$, $(-a)b = a(-b) = -(ab)$ and $(-a)(-b) = ab$.

E5.2 Find all units and all zero divisors of $\mathbb{Z}_{12}$.

E5.3 Show that $M_2(\mathbb{R})$ has zero divisors. Is it an integral domain?

E5.4 Prove that the units of $\mathbb{Z}[i]$ are exactly $\pm 1, \pm i$.

E5.5 A ring is *Boolean* if $a^2 = a$ for all a. Prove that a Boolean ring has characteristic 2 and is commutative.

E5.6 Let R be a commutative ring with unity. Call a *nilpotent* if $a^n = 0$ for some $n \geqslant 1$. Prove that the sum of two nilpotent elements is nilpotent.

E5.7 Let R be a commutative ring of prime characteristic p. Prove that $(a+b)^p = a^p + b^p$ for all $a, b \in R$.

E5.8 Prove that the characteristic of an integral domain is 0 or a prime.

More problems. Gallian: Ch. 12 and Ch. 13, the exercise sets on rings, integral domains and fields. Herstein: §3.1 and §3.2, the problem sets.

6 Ideals, Factor Rings, Prime and Maximal Ideals

6.1 Subrings, Ideals, and Ring Homomorphisms

Reading. Gallian, Ch. 14 (Ideals and Factor Rings) and Ch. 15 (Ring Homomorphisms); Herstein, §3.3 and §3.4.

Definition 6.1 (Subring)**.** A subset A of a ring R is a *subring* if A is itself a ring under the same operations.

Definition 6.2 (Ideal)**.** A subring A of a ring R is a (two-sided) *ideal* of R if for every $r \in R$ and every $a \in A$, both ra and ar are in A. In other words

$$rA = \{ra : a \in A\} \subseteq A \quad \text{and} \quad Ar = \{ar : a \in A\} \subseteq A \qquad \text{for all } r \in R.$$

We say that an ideal *absorbs* the elements of R. An ideal A is *proper* if $A \neq R$.

Remark 6.3. An ideal A is a normal subgroup of $(R, +)$, since $(R, +)$ is abelian.

Definition 6.4 (Ring homomorphism and kernel)**.** A map $\phi : R \to R'$ between rings is a *ring homomorphism* if $\phi(a+b) = \phi(a)+\phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in R$. Its *kernel* is $K_\phi = \{a \in R : \phi(a) = 0_{R'}\}$.

Lemma 6.5. *If ϕ is a ring homomorphism from R to R' with kernel K_ϕ, then*

(1) K_ϕ is a subgroup of R under addition;

(2) if $a \in K_\phi$ and $r \in R$, then ar and ra are in K_ϕ.

Hence K_ϕ is an ideal of R.

Proof. (1) ϕ is a group homomorphism from $(R,+)$ to $(R',+)$, so its kernel is a subgroup by Theorem 4.10.

(2) Let $a \in K_\phi$ and $r \in R$. Then $\phi(ar) = \phi(a)\phi(r) = 0_{R'} \cdot \phi(r) = 0_{R'}$. In the same way $\phi(ra) = \phi(r) \cdot 0_{R'} = 0_{R'}$. □

Definition 6.6 (Ideal generated by elements)**.** Let R be a commutative ring and $a_1, \ldots, a_n \in R$. Put

$$\langle a_1, \ldots, a_n \rangle = \{ r_1 a_1 + r_2 a_2 + \cdots + r_n a_n : r_1, \ldots, r_n \in R \}.$$

If R has unity and $n = 1$, then $\langle a \rangle = \{ra : r \in R\}$ is the *principal ideal* generated by a.

Lemma 6.7. *$\langle a_1, \ldots, a_n \rangle$ is an ideal of R.*

Proof. The sum of $\sum r_i a_i$ and $\sum r_i' a_i$ is $\sum (r_i + r_i') a_i$, and the negative of $\sum r_i a_i$ is $\sum (-r_i) a_i$. So the set is a subgroup of $(R,+)$. It is also closed under multiplication, because it is absorbing: $s \sum r_i a_i = \sum (s r_i) a_i$, and R is commutative so $ar = ra$. □

Example 6.8.

(1) $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$. Indeed $n\mathbb{Z} = \langle n \rangle = \{rn : r \in \mathbb{Z}\}$, which is an ideal by the lemma above.

(2) Let $\mathbb{R}[x]$ be the ring of polynomials with real coefficients, and let A be the set of polynomials with constant term 0. Then $A = \langle x \rangle$, so A is an ideal of $\mathbb{R}[x]$. Indeed, if $p \in A$ then $p(x) = a_1 x + \cdots + a_m x^m = x\,(a_1 + a_2 x + \cdots + a_m x^{m-1}) \in \langle x \rangle$, and every $x\,q(x)$ has constant term 0.

(3) Let $\mathbb{Z}[x]$ be the ring of polynomials with integer coefficients, and let I be the set of polynomials whose constant term is even. Then $I = \langle x, 2 \rangle$, and I is an ideal of $\mathbb{Z}[x]$. Indeed, if $p(0) = 2m$ then $p(x) = x\,q(x) + 2m$ for a polynomial q, so $p \in \langle x, 2 \rangle$. Conversely every element $x\,q(x) + 2s(x)$ of $\langle x, 2 \rangle$ has an even constant term.

6.2 Factor Rings

Reading. Gallian, Ch. 14 (Factor Rings); Herstein, §3.4.

Theorem 6.9 (Factor rings)**.** *Let R be a ring and let A be a subring of R. The set of cosets $R/A = \{r + A : r \in R\}$ is a ring under the operations*

$$(s + A) + (t + A) = s + t + A \qquad \text{and} \qquad (s + A)(t + A) = st + A$$

if and only if A is an ideal of R.

Proof. Addition on R/A is well-defined and makes R/A an abelian group, because A is a subgroup of the abelian group $(R,+)$. So everything depends on the multiplication. We need to show:

(a) If A is an ideal, then multiplication is well-defined and R/A is a ring.

(b) If multiplication is well-defined, then A is an ideal.

(a) Let A be an ideal. Suppose $s + A = s' + A$ and $t + A = t' + A$. Then $s' = s + a$ and $t' = t + b$ for some $a, b \in A$. So

$$s't' = (s + a)(t + b) = st + sb + at + ab.$$

Here $sb,\ at,\ ab \in A$ because A absorbs elements of R. So $s't' = st + c$ with $c = sb + at + ab \in A$, and therefore $s't' + A = st + A$. So multiplication is well-defined. The remaining ring axioms for R/A follow from the axioms in R:

$$\begin{aligned}
\big((r + A)(s + A)\big)(t + A) &= rst + A = (r + A)\big((s + A)(t + A)\big),\\
(r + A)\big((s + A) + (t + A)\big) &= r(s + t) + A = (rs + rt) + A = (r + A)(s + A) + (r + A)(t + A),\\
\big((s + A) + (t + A)\big)(r + A) &= (s + t)r + A = (sr + tr) + A = (s + A)(r + A) + (t + A)(r + A).
\end{aligned}$$

The zero element is $0 + A$ and the negative of $r + A$ is $-r + A$; the axioms for addition hold because $(R/A, +)$ is a quotient of an abelian group by a subgroup.

(b) Suppose multiplication is well-defined, and let $a \in A$, $r \in R$. Since $a \in A$ we have $a + A = 0 + A$. Because multiplication is well-defined,

$$(a + A)(r + A) = (0 + A)(r + A), \quad \text{that is,} \quad ar + A = 0 \cdot r + A = A.$$

So $ar \in A$. In the same way $(r + A)(a + A) = (r + A)(0 + A)$ gives $ra \in A$. So A is an ideal. □

Remark 6.10. To obtain a ring structure on the cosets we need a *two-sided* ideal.

Example 6.11 (Examples of factor rings)**.** (1) $\mathbb{Z}/4\mathbb{Z} = \{0 + 4\mathbb{Z},\ 1 + 4\mathbb{Z},\ 2 + 4\mathbb{Z},\ 3 + 4\mathbb{Z}\}$. For example

$$(2 + 4\mathbb{Z}) + (3 + 4\mathbb{Z}) = 5 + 4\mathbb{Z} = 1 + 4\mathbb{Z}, \qquad (2 + 4\mathbb{Z})(3 + 4\mathbb{Z}) = 6 + 4\mathbb{Z} = 2 + 4\mathbb{Z}.$$

This is arithmetic modulo 4.

(2) $2\mathbb{Z}/6\mathbb{Z} = \{0 + 6\mathbb{Z},\ 2 + 6\mathbb{Z},\ 4 + 6\mathbb{Z}\}$. Here the ring is $2\mathbb{Z}$ (the even integers) and $6\mathbb{Z}$ is an ideal of it. This is arithmetic modulo 6 on the even numbers.

(3) Let R be the ring of 2×2 matrices with integer entries, and let I be the set of matrices all of whose entries are even. Then I is an ideal: every entry of AB is a sum of products of an entry of A and an entry of B, and if $B \in I$ each product has an even factor, so $AB \in I$; similarly $BA \in I$. Two matrices lie in the same coset of I if and only if their difference has all entries even, that is, if and only if corresponding entries have the same parity. So

$$R/I = \left\{ \begin{bmatrix} r_1 & r_2 \\ r_3 & r_4 \end{bmatrix} + I : r_t \in \{0, 1\} \right\},$$

which has 16 elements. For example

$$\begin{bmatrix} 5 & 3 \\ 2 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 2 \\ 2 & -4 \end{bmatrix},$$

and the second matrix lies in I, so $\left[\begin{smallmatrix} 5 & 3 \\ 2 & -4 \end{smallmatrix}\right] + I = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right] + I$.

(4) Let $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ (the Gaussian integers) and consider $\mathbb{Z}[i]/\langle 2 - i\rangle$. In this ring $2 - i$ is zero, so i behaves like 2 in every coset. Precisely, $(a+bi)-(a+2b) = b(i-2) = -b(2-i) \in \langle 2-i\rangle$, so $a + bi + \langle 2 - i\rangle = a + 2b + \langle 2 - i\rangle$. Also $5 = (2 - i)(2 + i) \in \langle 2 - i\rangle$, so an integer m lies in the same coset as its remainder modulo 5. Hence every coset has a representative in $\{0, 1, 2, 3, 4\}$. For example $1 + 4i + 3i^2 = -2 + 4i$, and $-2 + 4i + \langle 2 - i\rangle = 6 + \langle 2 - i\rangle = 1 + \langle 2 - i\rangle$.

(5) $\mathbb{R}[x]/\langle x^2 + 1\rangle$. Here $\langle x^2 + 1\rangle = \{f(x)(x^2 + 1) : f \in \mathbb{R}[x]\}$. By division with remainder, every polynomial g can be written $g = q(x^2 + 1) + (ax + b)$ with $a, b \in \mathbb{R}$, so every coset has a representative $ax+b$. Two different polynomials $ax+b$ and $a'x+b'$ lie in different cosets, because their difference is non-zero of degree at most 1, so it is not a multiple of $x^2 + 1$. Hence

$$\mathbb{R}[x]/\langle x^2 + 1\rangle = \{\, ax + b + \langle x^2 + 1\rangle : a, b \in \mathbb{R}\,\}.$$

Since $x^2 \equiv -1$, the product is $(ax + b)(cx + d) \equiv (ad + bc)x + (bd - ac)$, which is the rule for multiplying $b + ai$ and $d + ci$. So the map $ax + b + \langle x^2 + 1\rangle \mapsto b + ai$ is a bijection that respects addition and multiplication, and $\mathbb{R}[x]/\langle x^2 + 1\rangle$ is the field $\mathbb{C}$ (Cauchy's construction of $\mathbb{C}$).

6.3 Prime Ideals and Maximal Ideals

Reading. Gallian, Ch. 14 (Prime and Maximal Ideals) and Ch. 16 (polynomial rings); Herstein, §3.5 and §3.9.

In this section R is a commutative ring with unity.

Definition 6.12 (Prime ideal)**.** A *prime ideal* of R is a proper ideal A of R such that if $a, b \in R$ and $ab \in A$, then $a \in A$ or $b \in A$.

Definition 6.13 (Maximal ideal)**.** A *maximal ideal* of R is a proper ideal A of R such that whenever B is an ideal of R with $A \subseteq B \subseteq R$, then $B = A$ or $B = R$.

Remark 6.14. A prime ideal captures "primeness" for ideals. A maximal ideal is one where you cannot fit any other ideal strictly between the ideal and R.

Example 6.15. Let p be a prime number and let $I = p\mathbb{Z}$, the multiples of p. If $ab \in I$, then $p \mid ab$, so $p \mid a$ or $p \mid b$ because p is prime. So I is a prime ideal of $\mathbb{Z}$.

Proposition 6.16 (Ideals of polynomial rings)**.** *Let F be a field and let $f \in F[x]$ be a non-constant polynomial.*

(1) Every non-zero ideal I of $F[x]$ equals $\langle g\rangle$, where g is a non-zero polynomial of least degree in I.

(2) $\langle f\rangle$ is a prime ideal if and only if f is irreducible (that is, f is not a product of two polynomials of smaller degree).

Proof. (1) Let $g \in I$ be non-zero of least degree. Clearly $\langle g\rangle \subseteq I$. Let $h \in I$. By division with remainder (possible since F is a field) $h = qg + r$ with $r = 0$ or $\deg r < \deg g$. Then $r = h - qg \in I$, so $r = 0$ by the choice of g. Hence $h = qg \in \langle g\rangle$.

(2) Suppose $f = gh$ with $\deg g, \deg h < \deg f$. Then $gh \in \langle f\rangle$. But a non-zero multiple of f has degree at least $\deg f$, so $g \notin \langle f\rangle$ and $h \notin \langle f\rangle$. So $\langle f\rangle$ is not prime. (It is proper because f is not a unit, being non-constant.)

Conversely, suppose f is irreducible and $ab \in \langle f\rangle$ with $a \notin \langle f\rangle$. Let $J = \{uf+va : u, v \in F[x]\} = \langle f, a\rangle$, an ideal containing f. By (1), $J = \langle d\rangle$ for some d. Since $f \in J$, $f = dm$ for some m. As f is irreducible, $\deg d = 0$ or $\deg m = 0$. If $\deg m = 0$, then m is a non-zero constant and $d = m^{-1}f$, so $a \in J = \langle f\rangle$, a contradiction. So d is a non-zero constant, hence a unit, and $1 \in J$ by Lemma 6.17. Write $1 = uf + va$. Then $b = ufb + vab \in \langle f\rangle$, because $ab \in \langle f\rangle$. So $\langle f\rangle$ is prime. □

Lemma 6.17. *If an ideal* B *of* R *contains a unit (in particular, if* $1 \in B$*), then* $B = R$*. So a proper ideal contains no unit.*

Proof. Let $u \in B$ be a unit. Then $1 = u^{-1}u \in B$ because B absorbs elements of R. For any $r \in R$ we then have $r = r \cdot 1 \in B$. So $B = R$. □

Theorem 6.18. *Let* A *be an ideal of* R*. Then* R/A *is an integral domain if and only if* A *is a prime ideal.*

Proof. R/A is a commutative ring with unity $1 + A$, whose zero is the coset $0 + A = A$.

($\Rightarrow$) Suppose R/A is an integral domain. Then $1 + A \neq 0 + A$, so $1 \notin A$ and $A \neq R$: the ideal is proper. Let $ab \in A$. Then $(a + A)(b + A) = ab + A = 0 + A$. Since R/A has no zero divisors, $a + A = 0 + A$ or $b + A = 0 + A$, that is, $a \in A$ or $b \in A$. So A is prime.

($\Leftarrow$) Suppose A is prime. Then A is proper, so $1 \notin A$ by Lemma 6.17, and $1 + A \neq 0 + A$. Suppose $(a + A)(b + A) = 0 + A$. Then $ab \in A$, so $a \in A$ or $b \in A$ because A is prime. So one of the two cosets is the zero coset. Hence R/A has no zero divisors and is an integral domain. □

Theorem 6.19. *Let* A *be an ideal of* R*. Then* R/A *is a field if and only if* A *is a maximal ideal.*

Proof. ($\Rightarrow$) Suppose R/A is a field. Then $1 + A \neq 0 + A$, so A is proper. Let B be an ideal with $A \subsetneq B$. Choose $b \in B$ with $b \notin A$. Then $b + A$ is a non-zero element of R/A, so it has an inverse: there is $c \in R$ with $(b + A)(c + A) = 1 + A$. This means $bc + A = 1 + A$, so $1 - bc \in A \subseteq B$. Also $bc \in B$, since $b \in B$ and B absorbs. Therefore

$$1 = (1 - bc) + bc \in B.$$

By Lemma 6.17, $B = R$. So A is maximal.

($\Leftarrow$) Suppose A is maximal. R/A is a commutative ring with unity $1 + A$, and $1 + A \neq 0 + A$ because A is proper. We must show that every non-zero element of R/A has an inverse. Let $b \in R$ with $b \notin A$, so that $b + A \neq 0 + A$. Put

$$B = \{\, br + a : r \in R,\ a \in A \,\}.$$

B *is an ideal.* It is closed under addition and negatives, since $(br + a) + (br' + a') = b(r + r') + (a + a')$ and $-(br + a) = b(-r) + (-a)$. It absorbs elements of R, since $s(br + a) = b(sr) + sa$ with $sa \in A$.

We have $A \subseteq B$ (take $r = 0$) and $b \in B$ (take $r = 1$, $a = 0$), while $b \notin A$. So $A \subsetneq B$. Because A is maximal, $B = R$. In particular $1 \in B$: there are $c \in R$ and $a' \in A$ with $1 = bc + a'$. Then $bc + A = 1 - a' + A = 1 + A$, that is,

$$(b + A)(c + A) = 1 + A.$$

So $b + A$ is a unit. Hence R/A is a field. □

Corollary 6.20. *In a commutative ring with unity, every maximal ideal is a prime ideal.*

Proof. Let A be maximal. By Theorem 6.19, R/A is a field. A field has no zero divisors: if $xy = 0$ and $x \neq 0$ then $y = x^{-1}xy = 0$. So R/A is an integral domain, and A is prime by Theorem 6.18. □

Example 6.21. (1) For a prime p, $\mathbb{Z}/p\mathbb{Z} = \mathbb{Z}_p$ is a field (Corollary 5.13), so $p\mathbb{Z}$ is a maximal ideal of $\mathbb{Z}$.

(2) $\mathbb{Z}/4\mathbb{Z}$ is not an integral domain, because $(2 + 4\mathbb{Z})(2 + 4\mathbb{Z}) = 0 + 4\mathbb{Z}$. So $4\mathbb{Z}$ is not a prime ideal of $\mathbb{Z}$. (Indeed $2 \cdot 2 \in 4\mathbb{Z}$ but $2 \notin 4\mathbb{Z}$.)

(3) $\mathbb{R}[x]/\langle x^2 + 1\rangle \cong \mathbb{C}$ is a field, so $\langle x^2 + 1\rangle$ is a maximal ideal of $\mathbb{R}[x]$.

Exercises

E6.1 Prove that every ideal of $\mathbb{Z}$ has the form $n\mathbb{Z}$ for some $n \geqslant 0$.

E6.2 Let A, B be ideals of R. Prove that $A \cap B$ and $A + B = \{a + b : a \in A, b \in B\}$ are ideals.

E6.3 Let R be commutative with unity. Prove that an ideal containing a unit is R. Prove that R is a field if and only if its only ideals are $\{0\}$ and R.

E6.4 Prove that a maximal ideal of a commutative ring with unity is prime.

E6.5 In $\mathbb{Z}[x]$ prove that $\langle x \rangle$ is prime but not maximal. (Hint: $\langle x \rangle \subsetneq \langle x, 2 \rangle \subsetneq \mathbb{Z}[x]$.)

E6.6 List all ideals of $\mathbb{Z}_{12}$. Which are prime? Which are maximal?

E6.7 Let $\varphi : R \to S$ be a ring homomorphism. Prove that $\operatorname{Ker} \varphi$ is an ideal of R, and that $\varphi^{-1}(B)$ is an ideal of R for every ideal B of S; it contains $\operatorname{Ker} \varphi$.

E6.8 Prove that $\langle 2, x \rangle$ is not a principal ideal of $\mathbb{Z}[x]$.

More problems. Gallian: Ch. 14, Ch. 15 and Ch. 16, the exercise sets on ideals, factor rings, ring homomorphisms and polynomial rings. Herstein: §3.3–§3.5 and §3.9, the problem sets.